Diabetes Treatment Algorithm — Instructions

Read this page before using the tool. Tabs are listed in the order you'll typically use them.

Instructions for Patient Information Worksheet

1	Enter all patient parameters in the fields outlined in red (age, gender, weight, height, serum creatinine).
2	Select Yes/No for each Medical History item, including the new Obesity/overweight flag — these drive the preferred-therapy recommendations on the DM Medications tab.
3	Calculated values (Ideal/Adjusted Body Weight, BMI, CrCl, eGFR) populate automatically in blue — no need to enter these.

Instructions for DM Medications Worksheet

1	Use the cardiorenal risk / comorbidity table to identify preferred therapy based on the history entered on the Patient Information tab (ASCVD, heart failure, CKD, obesity, MASLD/MASH).
2	Select an oral and/or injectable drug of choice from the red dropdown fields; dosing populates automatically in blue.
3	Reassess therapy, including the need for hypoglycemia-prone agents, every 3-6 months.

Instructions for Insulin Worksheet

1	Use the initial dosing section to calculate starting insulin doses from patient weight (pulled automatically from Patient Information).
2	Use the Adjusting Insulin Regimen section to calculate correction dose and insulin-to-carb ratio from entered blood glucose values.
3	Use the Insulin Conversions section when switching between insulin products, including the new once-weekly insulin icodec (Awiqli) calculator.

Instructions for SMBG Worksheet

1	Enter blood glucose readings in the red grid for each meal time (before/after breakfast, lunch, dinner, and bedtime).
2	Running averages, hypoglycemia count, and hyperglycemia count calculate automatically — hypoglycemic readings (<70) highlight red, readings over 180 highlight purple.
3	Reference current ADA blood sugar goals in the panel on the right, including the 2026 age/frailty-individualized and perioperative targets noted at the bottom of the tab.

This tool is intended to support, not replace, clinical judgment. Verify all calculated values and dosing recommendations — especially any field marked new for ADA 2026 — before use in patient care.

Created by	Andrew Tran, PharmD, BCACP, BCPS
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Created on	06/01/2019
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Diabetes Treatment Algorithm — Patient Information

Red = enter data here | Blue = calculated automatically | Green = updated for ADA Standards of Care in Diabetes—2026

Disclaimer: use of this tool does not replace clinical judgment.

Input Patient Parameters	
Patient Age (years)	60
Gender (male, female)	male
Total Body Weight (kg)	60
Height (in)	65
Ideal Body Weight (kg)	61.50
Adjusted Body Weight (kg)	60.90
Patient's BMI (%)	22.01
Weight used in CrCl calculation	Actual Body Weight
Serum Creatinine (mg/L)	1.2
CrCl via Cockcroft-Gault Equation (mL/min)	55.56
EGFR (mL/min)	65.33

Conversions	Enter	Conversion
Enter lbs to convert to Kg		
Enter kg to convert to lbs		
Enter cm to inches		
Enter inches to cm		

Medical History (Yes/No — see DM Medications tab)	
ASCVD (established or high-risk)	
Heart Failure (preserved of reduced EF)	
Chronic Kidney Disease (eGFR < 60 mL/min and/or UAC > 30 mg/g)	
Obesity / overweight (BMI ≥ 27 kg/m² with comorbidity, or BMI ≥ 30 kg/m²)	

Source: American Diabetes Association. Standards of Care in Diabetes—2026. Diabetes Care. 2026;49(Suppl 1). Insulin dosing verified against FDA prescribing information (DailyMed), accessed Aug 2026.

Type 2 Diabetes Pharmacotherapy

Healthy lifestyle behaviors, self-management education/support, and social determinants of health should be considered in ALL patients.

First-line pharmacotherapy should be selected based upon patient-specific factors: glycemic management needs, cardiorenal risk, comorbidities, cost, and access. Consider combination pharmacotherapy at initiation if A1C is more than 1.5% above target goal. ADA 2026 update: in patients with established/high-risk ASCVD, heart failure, CKD, or obesity, consider an SGLT2 inhibitor and/or GLP-1 RA (or GIP/GLP-1 RA) at or near diagnosis — independent of metformin use or baseline A1C — rather than only as a later add-on.	Consider early insulin initiation with extreme hyperglycemia: - Blood glucose > 300 mg/dL - A1C > 10% - Signs of catabolism

Reassess and modify treatment every 3-6 months.

Select Therapy Based on Cardiorenal Risk / Comorbidity

Comorbidity / Risk Factor	Preferred Therapy
ASCVD (established or high risk)	GLP-1 RA or SGLT2i with proven CVD benefit. Consider initiating at or near diagnosis in high-risk patients, independent of metformin use (ADA 2026).
Heart Failure (HFrEF or HFpEF)	SGLT2i with HF benefit preferred (avoid TZDs; avoid saxagliptin). GLP-1 RA with HF benefit, or GIP/GLP-1 RA (tirzepatide), also supported — particularly for HFpEF (ADA 2026, based on SUMMIT trial data; guideline-based, not an FDA cardiovascular-outcome label).
Chronic Kidney Disease (eGFR < 60 mL/min and/or UACR > 30 mg/g)	On maximally tolerated ACEi/ARB; SGLT2i with renal benefit. GLP-1 RA now also supported in T2DM with CKD (ADA 2026). Consider adding finerenone (nonsteroidal MRA) for persistent albuminuria (UACR ≥ 100 mg/g, eGFR 30-90) as a three-drug cardiorenal approach (ADA 2026).
Obesity / overweight (BMI ≥ 27 kg/m² with comorbidity, or BMI ≥ 30 kg/m²)	GLP-1 RA or GIP/GLP-1 RA (tirzepatide) preferred — prioritize the agent with the greatest weight-loss efficacy the patient can access and tolerate (ADA 2026 expanded indication branch).

MASLD / MASH (metabolic dysfunction-associated steatotic liver disease / steatohepatitis)	GLP-1 RA or GIP/GLP-1 RA preferred. Pioglitazone also has evidence in MASH but weigh against weight-gain and heart-failure risk (ADA 2026 expanded indication branch).

New for ADA 2026: reassess the need for, and dose of, hypoglycemia-prone agents (sulfonylureas, meglitinides, insulin) whenever a new glucose-lowering medication is added, to reduce hypoglycemia risk and treatment burden.

ORALS

Oral Drug of Choice?	**saxagliptin (Onglyza)**
Oral Drug Dosing	**2.5 mg or 5 mg PO daily**
How Supplied	
Renal Dosing	

INJECTABLES

Injectable Drug of Choice?	**dulaglutide (Trulicity)**
Injectable Drug Dosing	**Start 0.75 mg SQ once weekly; may increase to 1.5 mg weekly after at least 4 weeks; if additional control needed, may increase to 3 mg weekly, then 4.5 mg weekly (each after at least 4 weeks on prior dose); max 4.5 mg weekly**
How Supplied	
Renal Dosing	

FIRST-LINE THERAPY: METFORMIN

Dosage	- Initial, 500 mg orally twice daily OR 850 mg orally once daily with meals; may adjust dose in 500 mg increments weekly OR 850 mg every 2 weeks; if desired, patients initiated at 500 mg twice daily may be switched to 850 mg twice daily after 2 weeks; MAX dose, 2550 mg/day; max effective dose: 2000 mg daily. Doses greater than 2000 mg/day may be better tolerated if given in 3 divided doses - Renal impairment, eGFR below 30 mL/min/1.73 m(2): Use is contraindicated - Renal impairment, eGFR between 30 to 45 mL/min/1.73 m(2): Use not recommended - Renal impairment, eGFR falls below 45 mL/min/1.73 m(2) after initiation: Assess benefits and risks of continuing treatment - Renal impairment, eGFR falls below 30 mL/min/1.73 m(2) after initiation: Discontinue treatment

Recommended as foundational therapy independent of baseline A1C or A1C target (ADA 2026 — unchanged).

DRUG CLASS REFERENCE	
SGLT-2 Inhibitors	
CVD Benefit (FDA approved)	- canagliflozin (Invokana) - empagliflozin (Jardiance)
HF Benefit (FDA approved)	- dapagliflozin (Farxiga) - empagliflozin (Jardiance)
Renal benefit (FDA approved)	- canagliflozin (Invokana) - dapagliflozin (Farxiga) - empagliflozin (Jardiance)
Dosing	- canagliflozin (Invokana) - Start 100 mg PO daily; max 300 mg daily - dapagliflozin (Farxiga) - Start 5 mg PO daily; max 10 mg daily - empagliflozin (Jardiance) - Start 10 mg PO daily; max 25 mg daily
eGFR cutoffs (glycemic indication)	- canagliflozin (Invokana): eGFR 30-50: max dose 100 mg once daily; eGFR <30 or HD: Not recommended - dapagliflozin (Farxiga): eGFR < 60: Contraindicated - empagliflozin (Jardiance): eGFR < 45: Contraindicated
Cardiorenal combination (ADA 2026)	In T2DM with albuminuria (UACR ≥ 100 mg/g) and eGFR 30-90, consider combining an SGLT2i + finerenone (nonsteroidal MRA) + maximally tolerated ACEi/ARB for cardiorenal risk reduction.
GLP-1 RAs / GIP-GLP-1 RA	

CVD Benefit (FDA approved)	- dulaglutide (Trulicity) - liraglutide (Victoza) - semaglutide (Ozempic; Rybelsus)
HF Benefit	Neutral by FDA CV-outcome labeling for dulaglutide/liraglutide/semaglutide. GIP/GLP-1 RA (tirzepatide) shows benefit in HFpEF per the SUMMIT trial and is supported for HFpEF per ADA 2026 Standards of Care — this is guideline-based, not an FDA cardiovascular-outcome label.
Renal Benefit	Not FDA-approved as a dedicated renal-outcome indication for this class. ADA 2026 now supports GLP-1 RA / GIP-GLP-1 RA use in T2DM with CKD as part of individualized cardiorenal care.
Dosing	- dulaglutide (Trulicity) - Start 0.75 mg SQ once weekly; max to 1.5 mg SQ weekly - liraglutide (Victoza) - Start 0.6 mg SQ once daily for 1 week, then increase 1.2 mg once daily; then may increase to 1.8 mg daily - semaglutide (Ozempic) - Start 0.25 mg SQ once weekly for 4 weeks, then increase to 0.5 mg once weekly; may increase to 1 mg once weekly. If additional glycemic control needed after at least 4 weeks on 1 mg dosage, may increase to MAX of 2 mg once weekly - semaglutide (Rybelsus) - Start 3 mg orally daily for 30 days, then increase dose to 7 mg once daily, dose may be increased to 14 mg once daily after at least 30 days on 7 mg.

ADDITIONAL INJECTABLE DOSING REFERENCE (new — completes the drug list above)	
tirzepatide (Mounjaro)	Start 2.5 mg SQ once weekly x4 weeks (not for glycemic control); increase to 5 mg weekly; may increase in 2.5 mg increments after at least 4 weeks on current dose; max 15 mg weekly.
exenatide (Byetta) / exenatide ER (Bydureon)	Byetta: start 5 mcg SQ twice daily within 60 minutes before meals, may increase to 10 mcg twice daily after 1 month. Bydureon: 2 mg SQ once weekly, any time of day, with or without meals.
pramlintide (SymlinPen)	Adjunct to mealtime insulin in T2DM: start 60 mcg SQ immediately before major meals; may increase to 120 mcg as tolerated. Reduce prandial insulin doses by 50% when initiating to reduce hypoglycemia risk.

DEPRIORITIZED / FALLBACK AGENTS (ADA 2026)

Sulfonylureas, meglitinides, and DPP-4 inhibitors remain fallback / cost-driven options — they lack cardiovascular, kidney, weight, or liver benefit relative to preferred agents. Use sulfonylureas and TZDs judiciously at the lowest effective dose given weight-gain and fracture risk, particularly in older adults.

Source: American Diabetes Association. Standards of Care in Diabetes—2026. Diabetes Care. 2026;49(Suppl 1), Section 9 (Pharmacologic Approaches to Glycemic Treatment). Injectable dosing verified against FDA prescribing information (DailyMed), accessed Aug 2026. Verify current oral semaglutide (Rybelsus) dosing/branding against the live FDA label before relying on this tool clinically — trade press reported a possible reformulation/brand transition not independently confirmed here.

Diabetes Treatment Algorithm — Insulin Dosing

Select or Evaluate Initial Insulin Dose | Red = enter data | Blue = calculated | Green = new for ADA 2026

Disclaimer: use of this tool does not replace clinical judgment.

Select Insulin Dosing Regimen — Initial

Initiating Insulin in Type 1 Diabetes

Insulin TDD	0.4-1.0 units/kg/day		
	24 - 60 units q daily		

Basal-Bolus Strategy

If using Long-Acting + Rapid-Acting insulin (50/50)	**12 - 30 units of basal q daily**	*PLUS*	**4 - 10 units of bolus TID with meals**
If using NPH and regular insulin (2/3 , 1/3)	**16 - 40 units of NPH BID**	*PLUS*	**8 - 20 units of regular BID with meals**
if using insulin 70/30 (0.5 units/kg/day -- > 2/3 AM + 1/3 PM)	**20 units of 70/30 AM + 10 units of 70/30 PM with meals**		

Initiating Insulin in Type 2 Diabetes

Basal Insulin Dose	Basal Insulin is initated at 10 units daily OR 0.1-0.2 units/kg/day		
	10 units q daily	*OR*	**6 - 12 units q daily**

Adjusting Insulin Regimen

Enter TDD of insulin			
Adding bolus insulin in T2DM	If not at goal, bolus insulin is initiated at 4 units/day OR 10% of TDD		
	4 units per day with largest meal	*OR*	
Enter rapid or regular Insulin as bolus?			
Insulin sensitivity factor (rule of 1800 or 1500)			
Insulin-to-carb ratio (rule of 500 or 450)			
Correction Dose	[Current BG - Target BG]/Correction Factor		
Enter current BG			
Enter target BG			
Calculated Correction Dose			

Insulin Conversions

Basal insulin conversions	
Enter current basal insulin	
Enter current insulin dose	
Enter desired basal insulin	
Calculated desired insulin dose (TDD)	
Bolus insulin conversions	
Enter current bolus insulin	
Enter current insulin dose	
Enter desired bolus insulin	
Calculated desired insulin dose	
Mixed insulin conversions	
Enter current mixed insulin	
Enter current insulin dose	
Enter desired mixed insulin	
Calculated desired insulin dose (TDD)	

ONCE-WEEKLY BASAL INSULIN — INSULIN ICODEC (AWIQLI) — new, FDA-approved March 2026

Formulation	U-700 (700 units/mL) — about 7x more concentrated than standard U-100 basal insulins. Use only the Awiqli FlexTouch pen; never withdraw into a syringe. Do not mix or dilute with any other insulin.
Insulin-naive starting dose	70 units SQ once weekly, same day each week.
Enter current total daily dose of basal insulin (units) — for patients switching from daily basal	
Week 1 dose (one-time starting dose)	
Week 2 dose	
Week 3 and beyond	Titrate from the prior weekly dose based on metabolic needs, blood glucose monitoring, and glycemic goals.

Administration notes	Give the first Awiqli dose the day after the last daily basal dose. Doses in 10-unit increments; max 700 units per single injection (doses above 700 units require split injections). If a dose is missed, take within 4 days of the missed dose; if more than 4 days have passed, skip and resume on the next scheduled day.

Source: Awiqli (insulin icodec-abae) FDA Prescribing Information, DailyMed, revised 3/2026, accessed Aug 2026. All other insulin initiation/titration figures on this tab are carried over from prior Standards of Care — no ADA-specific numeric changes were identified for 2026.

Diabetes Treatment Algorithm — SMBG Averages

SMBG Averages | Red = enter data | Blue = calculated | Green = new for ADA 2026

Disclaimer: use of this tool does not replace clinical judgment.

	Enter SMBG values						Running average
Before Breakfast							
After Breakfast							
Before Lunch							
After Lunch							
Before Dinner							
After Dinner							
Bedtime							

ADA Blood Sugar Goals	
A1c	<7%
Fasting	80-130
2 hours post-prandial	< 180
Hypoglycemia	<70

Hypoglycemic SMBG < 70 will change cell red

SMBG > 180 will change cell purple

Overall Average	
# Hypoglycemic SMBG	**0**
# SMBG > 180	**0**

Conversions	Enter	Conversion
A1C to estimated glucose		
Average glucose to A1c		

ADDITIONAL 2026 ADA TARGETS — new

Glycemic and CGM targets are now framed more explicitly around age, frailty, and functional/health status rather than a single flat number — individualize the <7% A1C goal accordingly, especially in older adults (ADA 2026, Sections 6 and 13).

New perioperative target (ADA 2026): A1C ≤ 8% before elective surgery; inpatient glucose 100-180 mg/dL before, during, and after surgery.

Source: American Diabetes Association. Standards of Care in Diabetes—2026. Diabetes Care. 2026;49(Suppl 1), Sections 6 and 13. Core fasting/postprandial/hypoglycemia numeric thresholds (80-130, <180, <70 mg/dL) verified as carryover with no confirmed 2026 numeric change; cross-check Table 6.2 directly if this tool is used for a non-standard target population.